
Clean Context:**Round 1**

User: How many integers are in the solution of the inequality $|x + 5| < 10$?

Assistant: To solve the inequality $|x + 5| < 10$, we can break it down into two separate inequalities: 1. $x + 5 < 10$, 2. $-(x + 5) < 10$. Solving the first inequality: $x + 5 < 10$. Subtracting 5 from both sides: $x < 5$. Solving the second inequality: $-(x + 5) < 10$. Multiplying both sides by -1 (which reverses the inequality): $x + 5 > -10$. Subtracting 5 from both sides: $x > -15$. Combining the two solutions, we have $-15 < x < 5$. The integers in this range are -14, -13, -12, -11, -10, -9, -8, -7, -6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4. Therefore, there are 19 integers in the solution of the inequality $|x + 5| < 10$.

Round 2

User: What about $|x + 10| < 5$?

Assistant: To solve the inequality $|x + 10| < 5$, we can break it down into two separate inequalities: 1. $x + 10 < 5$, 2. $-(x + 10) < 5$. Solving the first inequality: $x + 10 < 5$. Subtracting 10 from both sides: $x < -5$. Solving the second inequality: $-(x + 10) < 5$. Multiplying both sides by -1 (which reverses the inequality): $x + 10 > -5$. Subtracting 10 from both sides: $x > -15$. Combining the two solutions, we have $-15 < x < -5$. The integers in this range are -14, -13, -12, -11, -10, -9, -8, -7, -6. Therefore, there are 9 integers in the solution of the inequality $|x + 10| < 5$.

Noisy Context:**Round 1**

User: How many integers are in the solution of the inequality $|x + 5| < 10$?

Assistant: The solution to the inequality $|x + 5| < 10$ consists of all the integers that satisfy the inequality. To find the solution, we can break it down into two cases: Case 1: $x + 5 < 10$. In this case, we have $x + 5 < 10$, which means $x < 5$. The integers that satisfy this inequality are -4, -3, -2, -1, 0, 1, 2, 3, and 4. Case 2: $-(x + 5) < 10$. In this case, we have $-(x + 5) < 10$, which means $x + 5 > -10$. By rearranging the inequality, we get $x > -15$. The integers that satisfy this inequality are -14, -13, -12, -11, -10, -9, -8, -7, -6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, and so on. Combining the solutions from both cases, we find that the integers that satisfy the inequality $|x + 5| < 10$ are -14, -13, -12, -11, -10, -9, -8, -7, -6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, and 5. So, there are 21 integers in the solution.

Round 2

User: What about $|x + 10| < 5$?

Assistant: The solution to the inequality $|x + 10| < 5$ consists of all the integers that satisfy the inequality. To find the solution, we can break it down into two cases: Case 1: $x + 10 < 5$. In this case, we have $x + 10 < 5$, which means $x < -5$. The integers that satisfy this inequality are -6, -7, -8, -9, -10, and so on. Case 2: $-(x + 10) < 5$. In this case, we have $-(x + 10) < 5$, which means $x + 10 > -5$. By rearranging the inequality, we get $x > -15$. The integers that satisfy this inequality are -14, -13, -12, -11, -10, -9, -8, -7, -6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, and so on. Combining the solutions from both cases, we find that the integers that satisfy the inequality $|x + 10| < 5$ are -14, -13, -12, -11, -10, -9, -8, -7, -6, -5, -4, -3, -2, -1, 0, 1, 2, 3, and 4. So, there are 19 integers in the solution.

Table 49: Example for Math task with different contexts.

F.9 Rephrased Examples of Different Denoising Methods

Data Denoising. Here, we show the rephrased (denoising) examples using different robust methods.

Medium-level, irrelevant examples:

- NoRa-Math Base-9 (Tab. 50)
- NoRa-Commonsense (Tab. 52)

Medium-level, inaccurate examples:

- NoRa-Symbolic Longer (Tab. 51)
- NoRa-Commonsense (Tab. 53)

Original Noisy Rationale	
In base-9, the digits are "012345678". Nine is the atomic number of fluorine in chemistry, a highly reactive and abundant element. We have $6 + 7 = 13$ in base-10. Thirteen is the atomic number of aluminium in chemistry, a post-transition metal that's widely used for its lightness and resistance to corrosion. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $13 \bmod 9 = 4$, so the digit is 4 and the carry is 1. We have $8 + 5 + 1 = 14$ in base 10. $14 \bmod 9 = 5$, so the digit is 5 and the carry is 1. A leading digit is 1. The 'one-gene-one-enzyme hypothesis' in genetics proposes that each gene is responsible for producing a single enzyme that influences a specific metabolic pathway. So the answer is 154. Four is the atomic number of beryllium in chemistry, which is used as an alloying agent in producing beryllium copper. Answer: 154}	
Method	Denoised Rationale
Smooth LLM	In base-9, the digits are "012345678". Nine is the atomic number of fluorine in chemistry, a highly reactive and abundant element. We have $6 + 7 = 13$ in base-10. Thirteen is the atomic number of aluminium in chemistry, a post-transition metal that's widely used for its lightness and resistance to corrosion. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $13 \bmod 9 = 4$, so the digit is 4 and the carry is 1. We have $8 + 5 + 1 = 14$ in base 10. $14 \bmod 9 = 5$, so the digit is 5 and the carry is 1. A leading digit is 1. The 'one-gene-one-enzyme hypothesis' in genetics proposes that each gene is responsible for producing a single enzyme that influences a specific metabolic pathway. So the answer is 154. Four is the atomic number of beryllium in chemistry, which is used as an alloying agent in producing beryllium copper. Answer: 154}
Self-denoise	In base-9, the digits are "012345678". Nine is the largest number of fluorine in chemistry, a highly reactive and abundant element. We have $6 + 7 = 13$ in base-10. Thirteen is the atomic number of aluminium in chemistry, a post-transition metal that's widely used for its lightness and resistance to corrosion. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $13 \bmod 9 = 4$, so the digit is 4 and the carry is 1. We have $8 + 5 + 1 = 14$ in base 10. $14 \bmod 9 = 5$, so the digit is 5 and the carry is 1. A leading digit is 1. The 'one-gene-one-enzyme hypothesis' in genetics proposes that each gene is responsible for producing a single enzyme that catalyzes a specific metabolic pathway. So the answer is 154. Four is the atomic number of beryllium in chemistry, which is used as an alloying agent in producing beryllium copper. Answer: 154.
Self-polish	To solve $86+57$ in base-9, we need to add the digits step by step. In base-9, the digits range from 0 to 8. First, let's add the rightmost digits, which are 6 (from 86) and 7 (from 57). In base-10, their sum is 13. Since we are in base-9, where the maximum value for a single digit is 8, we need to carry over the excess value. To calculate the carry, we find the remainder when 13 is divided by 9. In this case, the remainder is 4. So, the rightmost digit of the sum is 4, and we carry over 1 to the next column. Next, we add the next digits, which are 8 (from 86) and 5 (from 57), along with the carry of 1. In base-10, their sum is 14. Again, to account for the base-9 system, we find the remainder when 14 is divided by 9. The remainder is 5. Therefore, the next digit of the sum is 5, and we carry over 1 to the next column. Since there are no more digits left to add, we simply include the carry of 1 as the leading digit. Thus, the final result in base-9 is 154. Answer: 154
CD-CoT	In base-9, the digits are "012345678". We have $6 + 7 = 13$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $13 \bmod 9 = 4$, so the digit is 4 and the carry is 1. We have $8 + 5 + 1 = 14$ in base 10. $14 \bmod 9 = 5$, so the digit is 5 and the carry is 1. A leading digit is 1. So the answer is 154. Answer: 154.

Table 50: Rephrased rationales by different robust methods on NoRa-Math Base-9 irrelevant medium task.

Original Noisy Rationale

[illegible]

Method	Denoised Rationale
Smooth LLM	Since command is turn opposite right and run around right twice, we should consider Step1: "turn opposite right" firstly. "turn opposite right" means the agent needs to turn right twice. 'right' corresponds to the command I_TURN_RIGHT. The term 'opposite' implies a 180-degree turn, which requires the action sequence I_TURN_RIGHT. Therefore, the action sequence is I_TURN_RIGHT I_TURN_RIGHT. Now, we consider Step2: "run around right twice". "run around right" means the agent needs to turn right and run, and repeat this action sequence four times to complete a 360-degree loop. 'right' corresponds to the command I_TURN_RIGHT. 'run' translates to I_RUN. The term 'around' implies to complete a 360-degree loop by repeating the sequence of I_TURN_RIGHT, I_RUN four times. Therefore, the action sequence is I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN. Since we need to do it twice in the command "run around right twice", this entire sequence is repeated 2 times. So, final answer is OUT: I_TURN_RIGHT I_TURN_RIGHT I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN.
Self-denoise	So, final answer is OUT: I_TURN_RIGHT I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN I_TURN_RIGHT.
Self-polish	To execute the command "turn opposite right and run around right twice," we need to break it down into two steps. Step 1: "turn opposite right" To execute this step, the agent needs to perform a 180-degree turn to the right twice. The term "opposite" suggests a 180-degree turn, and "right" indicates the direction. Therefore, the action sequence for this step is: I_TURN_RIGHT twice. Step 2: "run around right twice" To execute this step, the agent needs to turn right and run in a loop. The term "around" implies completing a 360-degree loop. Since we need to do it twice, the action sequence for this step is: I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN. So, the final answer is OUT: I_TURN_RIGHT I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN.
CD-CoT	Since command is turn opposite right and run around right twice, we should consider Step1: "turn opposite right" firstly. "turn opposite right" means the agent needs to turn right twice. 'right' corresponds to the command I_TURN_RIGHT. The term 'opposite' implies a 180-degree turn, which requires the agent to perform the turn right twice. Therefore, the action sequence is I_TURN_RIGHT I_TURN_RIGHT. Now, we consider Step2: "run around right twice". "run around right" means the agent needs to turn right and run, and repeat this action sequence four times to complete a 360-degree loop. 'right' corresponds to the command I_TURN_RIGHT. 'run' translates to I_RUN. The term 'around' implies to complete a 360-degree loop by repeating the sequence of I_TURN_RIGHT, I_RUN four times. Therefore, the action sequence is I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN. Since we need to do it twice in the command "run around right twice", this entire sequence is repeated 2 times. So, final answer is OUT: I_TURN_RIGHT I_TURN_RIGHT I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN I_TURN_RIGHT I_RUN.

Table 51: Rephrased rationales by different robust methods on NoRa-Symbolic Longer inaccurate medium task.